MS 8121: LINEAR ALGEBRA AND CALCULUS

SOME APPLICATIONS OF MATRICES

A) Combinatorial Problems: Graph Theory

Concepts:

1) Matrices play an important role in combinatorial problems of many different types and, in particular, in graph theory.

2) The purpose of the brief account offered here will be to illustrate a particular application of matrices, and no attempt will be made to discuss their subsequent use in the solution of the associated problems.

3) Combinatorial problems involve dealing with the possible arrangements of situations of various different kinds, and computing the number and properties of such arrangements.

4) The arrangements may be of very diverse type, involving at one extreme the ordering of matches that are to take place in a tennis tournament, and at the other extreme finding an optimum route for a delivery truck or for the most efficient routing of work through a machine shop.

5) The ideas involved are most easily illustrated by means of examples, the first of which involves the delivery from a storage depot of a consumable product to a group of supermarkets in a large city where it is important that daily deliveries be made as rapidly as possible. One possibility involves a delivery truck making a delivery to each supermarket in turn and returning to the storage depot between each delivery before setting out on the next delivery. An alternative is to travel between supermarkets after each delivery without returning to the storage depot.

6) The question that then arises is which approach to routing is the best, and how it is to be determined. A typical situation is illustrated in Figure 1 below, in which supermarkets numbered 1 to 5 are involved, with circles representing supermarkets and lines and arcs representing the routes.

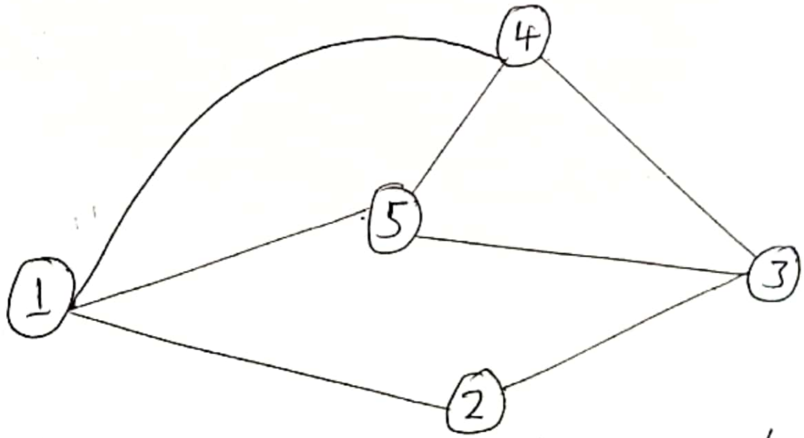


Figure 1: The graph representing routes

7) The representation in Figure 1 is called a Graph, and is to be regarded as a set of points represented by the circles called Vertices of the graph, and Edges of the graph represented by the lines and arcs.

8) In Figure 1, the vertices are the circles 1, 2, ..., 5 and the seven edges are the lines and arcs connecting the vertices.

9) A special type of matrix associated with such a graph is an Adjacency Matrix / Incidence Matrix, that is, a matrix whose only entries are 0 to 1.

10) The rules for the entries in an adjacency matrix $A = [a_{ij}]$ are that:

$$a_{ij} = \begin{cases} 1; & \text{if vertices } i \text{ and } j \text{ are joined by an edge} \\ 0; & \text{otherwise} \end{cases}$$

⇒ Thus, the adjacency matrix for the graph in Figure 1 above is seen to be the symmetric matrix.

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{bmatrix}$$

⇒ It is to be expected that an adjacency matrix is symmetric, because if i is adjacent to j, then j is adjacent to i.

11) Although we shall not attempt to do so here, the interconnection properties of the problem represented by the graph in Figure 1 can be analyzed in terms of its adjacency matrix. The optimum routing problem can then be resolved once the traveling times along roads (lines or arcs) are known.

12) Sometimes it happens that the edges in a graph represent connections that only operate in one direction, so then arrows must be added to the graph to indicate these directions. A graph of this type is called a <u>Digraph</u> (Directed Graph), as shown in Figure 2 below:

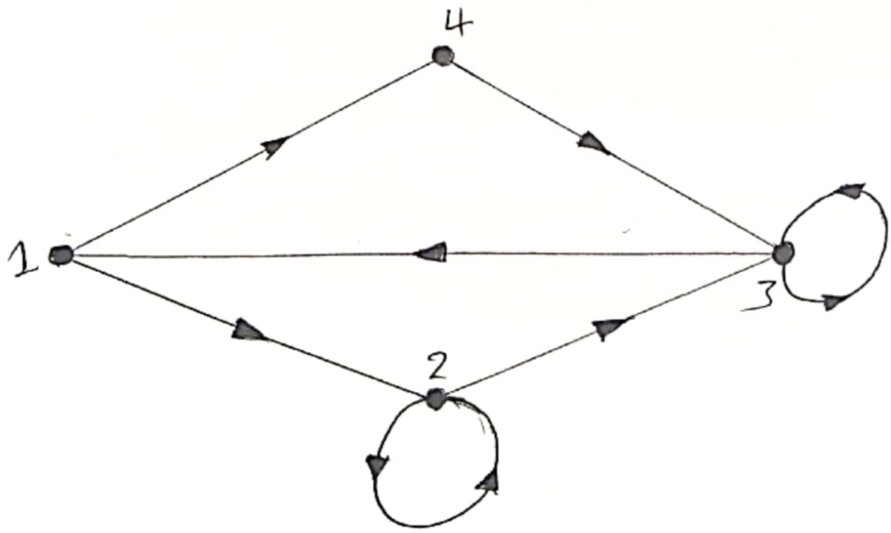


Figure 2: A typical digraph

13) The rules for the entries in the adjacency matrix $A = [a_{ij}]$ of a digraph are that

$$a_{ij} = \begin{cases} 1; & \text{if } i \text{ and } j \text{ are joined by an edge with an arrow from } i \text{ to } j \\ 0; & \text{otherwise.} \end{cases}$$

⇒ Thus, the associated adjacency matrix for the digraph in Figure 2 above is:

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

14) The adjacency matrix A characterizes all the possible interconnections between the four vertices and, as with the previous example, an analysis of the properties of any situation capable of representation in terms of this digraph can be performed using the matrix A. Problems of this type can arise in transportation problems in cities with one-way streets, and in chemical processes where a fluid is piped to different parts of a plant through an interconnecting network of pipes through which fluid may only flow in a given direction.

15) Internet search engines use matrices to keep track of the locations of information, the type of information at a location, keywords that appear in the information and even the way websites link to one another.

16) A large measure of the effectiveness of the search engine Google is the manner in which matrices are used to determine which sites are referenced by other sites. That is, instead of directly keeping track of the information content of an actual webpage or of an individual search topic, Google's matrix structure focuses on finding webpages that match the search topic, and then presents a list of such pages in the order of their importance.

17) Suppose that there are n accessible web pages during a certain month. A simple way to view a matrix that is part of Google's scheme is to imagine an $n \times n$ matrix A, called the "connectivity matrix", that initially contains all zeros. To build the connections, proceed as follows. When you detect that website j links to website i, set entry a_{ij} equal to one.

Since n is quite large, in the billions, most entries of the connectivity matrix A are zero. (Such a matrix is called sparse). If row i of A contains many ones, then there are many sites linking to site i. Sites that are linked to by many other sites are considered more important (or to have a higher rank) by the software driving the Google search engine. Such sites would appear near the top of a list returned by a Google search on topics related to the information on site i. Since Google updates its connectivity matrix about every month, n increases over time and new links and sites are adjoined to the connectivity matrix.

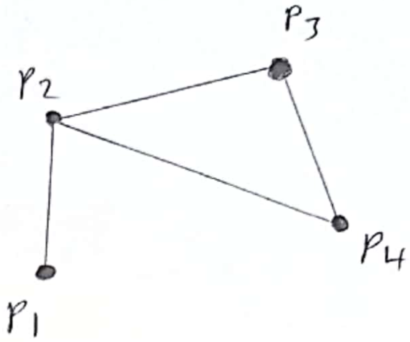
18) By a <u>graph</u> we mean a set of points called <u>nodes</u> or <u>vertices</u>, some of which are connected by lines called <u>Edges</u>.

19) The nodes/vertices are usually labeled as $P_1, P_2, \ldots, P_k$, and for now we allow an edge to be travelled in either direction.

20) One mathematical representation of a graph is constructed from a table.
⇒ For example, the following table represents the

graph shown.

	P_1	P_2	P_3	P_4
P_1	0	1	0	0
P_2	1	0	1	1
P_3	0	1	0	1
P_4	0	1	1	0



⇒ That is, the (i, j) entry $= 1$ if there is an edge connecting vertex P_i to vertex P_j; otherwise, the (i, j) entry $= 0$.

⇒ The incidence matrix A is the $k \times k$ matrix obtained by omitting the row and column labels from the preceding table. The incidence matrix for the corresponding graph is:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$